

# Lake Mead Water Conservation Accounting (ICS) and Storage

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## Overview

This is an R Markdown document. The document generates the following 3 plots:

1. ICS account balances by state and year (stacked bar)
2. Deposits and debits to ICS accounts by state and year (stacked bar)
3. Stacked line plot that show Lake Mead Storage, Water Conservation Account (ICS) Balance, and reservoir protection elevation over time. The plot also shows a line of the anticipated Lake Mead storage volume were the Water Conservation Program not exist (states continued withdraws at their historical allocations).

## Data wrangling strategy

1. Import the reservoir elevation-volume curves
2. Import the water conservation account data for Lake Mead
3. Import historical reservoir storage data
4. Turn annual ICS data into Monthly, Calculate the Lake Mead Elevation without ICS accounts, and turn reservoir, ICS, and protection volumes into dataframe in narrow format for plotting as a stacked line graph.

The plots support the analysis of the work: “Lake Mead Water Conservation Accounts: Examples, Successes, Challenges, and Potential Improvements post 2026.”

This is a beginning R-programming effort! There could be lurking bugs or basic coding errors that I am not even aware of. Please report bugs/feedback to me (contact info below)

## Requested Citation

David E. Rosenberg (2026), “Lake Mead Water Conservation Accounting (ICS) and Storage.” Utah State University. Logan, Utah. <https://github.com/dzeke/ColoradoRiverCollaborate/tree/main/HistoricalCurrentReservoirData>.

## Current Status

Total Deposits: NA million acre-feet

Total Withdraws: NA million acre-feet

Current Balance: 3.6 million acre-feet

## Summary by State and Country (million acre-feet)

| State/Country | TotalDeposits | TotalDebits | Balance |
|---------------|---------------|-------------|---------|
| Arizona       | 0.75          | 0.05        | 0.70    |

| State/Country | TotalDeposits | TotalDebits | Balance |
|---------------|---------------|-------------|---------|
| California    | 2.34          | 0.61        | 1.73    |
| Nevada        | 1.09          | 0.20        | 0.89    |
| Mexico        | NA            | NA          | 0.13    |

Figure 1: Lake Mead Water Conservation Account (ICS) Balances

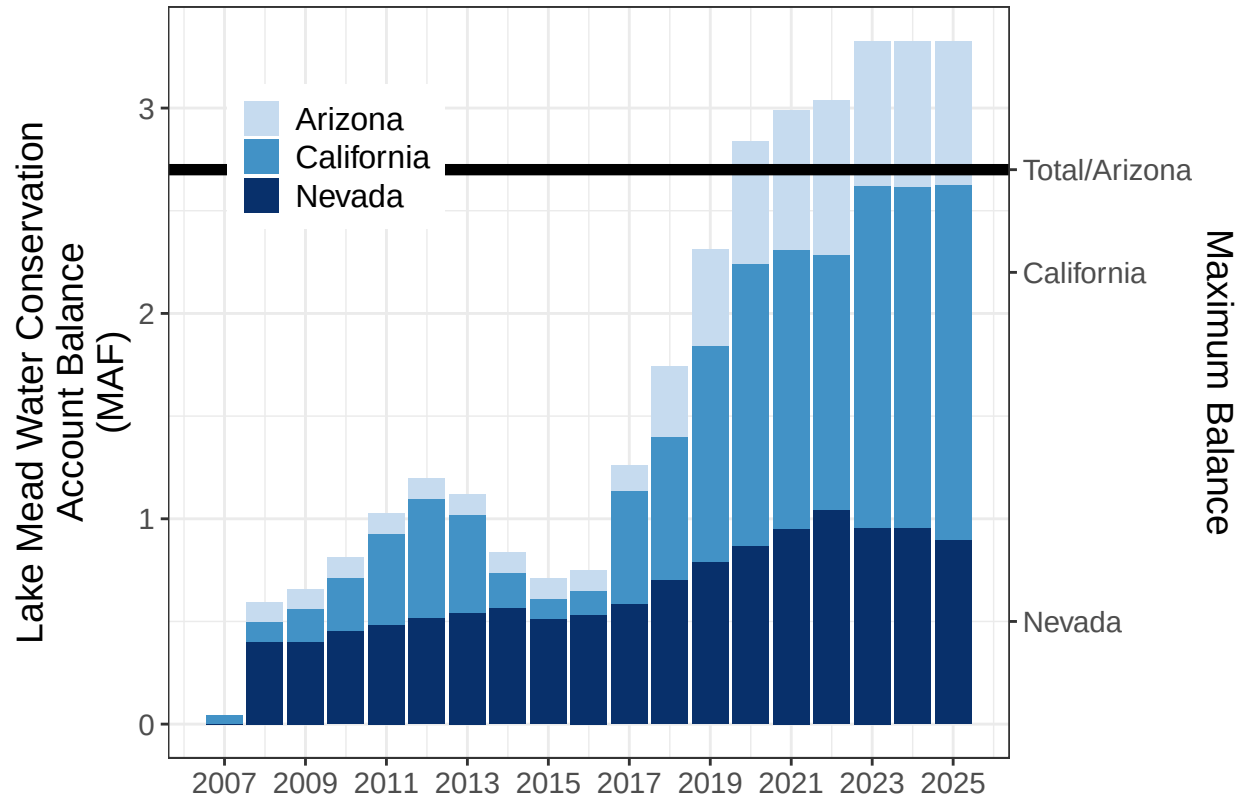


Figure 2. Lake Mead Water Conservation Account (ICS) Deposits and Withdraws

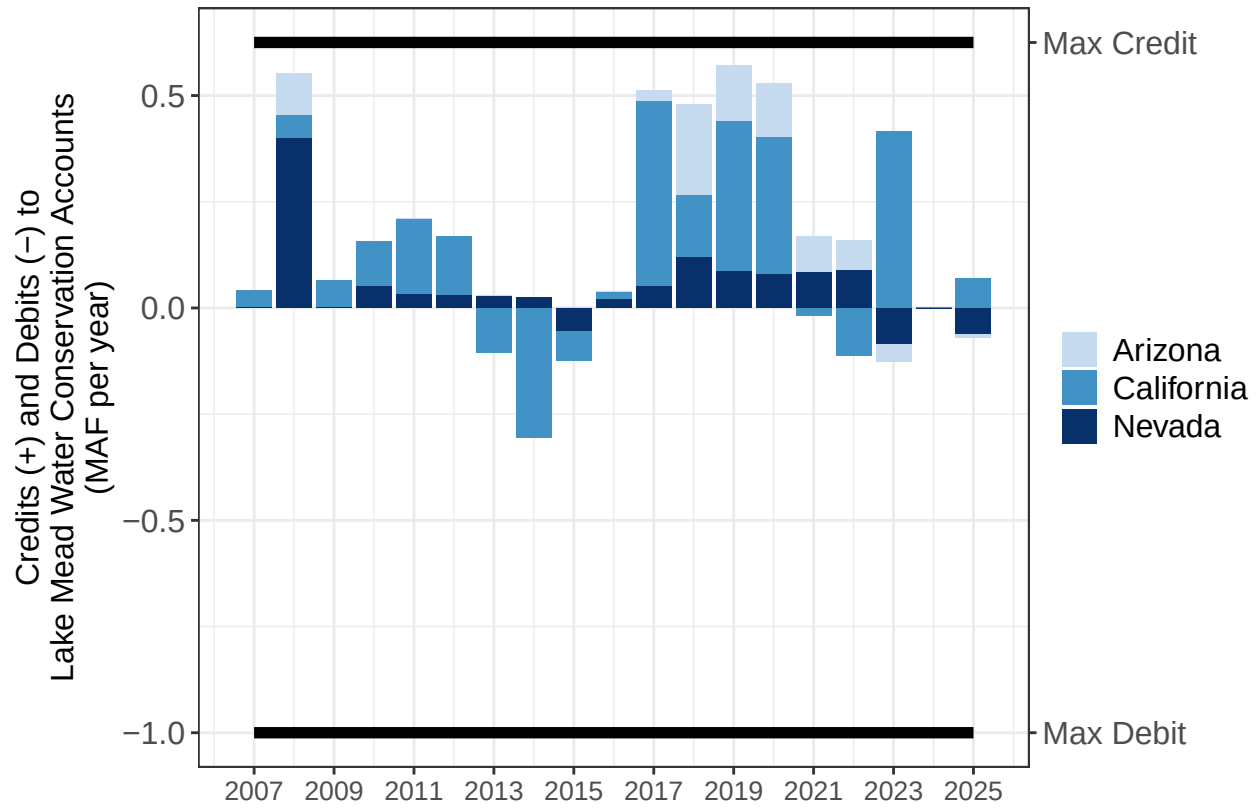


Figure 3. Lake Mead Storage (black), Water Conservation Account (ICS) Balances (light blue), Storage without the Water Conservation Program (dashed red), and Protection Volume (dark blue) Over Time.

